

capacity of one as due to interference issues, the law of diminishing returns is at play. Moreover, doubling the infrastructure will not double the revenue. Spectral efficiency has also improved over the years, but recently the increase in wireless spectral efficiencies has slowed to just 12 per cent year-on-year. So in the long run, what is the way out?

Dr Povey says that wireless optical systems can break all three of these restrictions. There is plenty of bandwidth. Modulation or demodulation is direct as there are no radios or antennae. Many nodes are feasible and they do not interfere like radio does. With plenty of resources, spectral efficiency is less sensitive and what has been developed for radio spectral efficiency can be applied to optical (with some minor tweaks). In short, Dr Povey has laid before us the case for visible light communications (VLC).

LED lights can be switched off and switched on faster than we can perceive. This on-off motion can be used to represent 0's and 1's—in other words, digital information. A sequence of such variations can cause a flow of data. This, or transmission by more complex modulation schemes such as Orthogonal Frequency Division Multiplexing (OFDM), is the technological basis of visible light communication or wireless optical communication.

Hello Li-Fi!

Li-Fi is a VLC technology developed by a team of scientists including Dr Gordon Povey, Prof. Harald Haas and Dr Mostafa Afgani at the University of Edinburgh. The term Li-Fi was coined by Prof. Haas when he amazed people by streaming high-definition video from a standard LED lamp, at *TED Global* in July 2011. Li-Fi is now part of the Visible Light Communications (VLC) PAN IEEE 802.15.7 standard.

Graham Bell tried this much earlier!

Alexander Graham Bell had experimented with visible light communications way back in 1880. He sent a voice message by modulating light. A parabolic mirror captured and projected his vocal vibrations. The drawback was that he used sunlight, which is highly unpredictable in nature. Today, the D-Light project has done something similar, using the more reliable LED technology!

How optical communication solves bandwidth crunch

For India, since the existing broadband capacity is extremely limited, it is expected that the demand will grow at an exponential rate in the foreseeable future. To meet this demand, a network of networks and a range of technology options need to be implemented. Many of these networks complement each other.

Wi-Fi is a very good complement to cellular network and has become quite pervasive globally as a local-area network (LAN). However, since the most critical bottleneck in any wireless-based network is the air access interface, Wi-Fi is unlikely to keep up with the ever-increasing demand for bandwidth. Clearly, the bandwidth provided to the end customer is dependent upon the spectrum efficiency (bits/second per hertz of frequency spectrum bandwidth) among other limitations.

The frequency of operation and the bandwidth available for Wi-Fi networks result in the requirement for additional options. Various options include increasing the frequency of operation and associated spectrum, spatial diversity (multiple antennae), cells to cover smaller and smaller geographic ranges, cell splitting, modulation and multiple access techniques.

Clearly, there are basic technological shortcomings in Wi-Fi, e.g., security issues and sub-optimal multiple access techniques, but it is the physical nature of the air interface which requires one to look for additional solutions. Alternatives like femto-cells do augment the capacity via their operation at a different spectrum, but a key approach is to supplement LAN coverage of the Wi-Fi network via use of personal or proximity area networks (PANs).

—Dr Suresh Borkar, a faculty member at the Electrical and Computer Engineering Department, Illinois Institute of Technology

“Li-Fi is typically implemented using white LED light bulbs. These devices are normally used for illumination by applying a constant current through the LED. However, by fast and subtle variations of the current, the optical output can be made to vary at extremely high speeds. Unseen by the human eye, this variation is used to carry high-speed data,” says Dr Povey.

In simple terms, Li-Fi can be thought of as a light-based Wi-Fi. That is, it uses light instead of radio waves to transmit information. And instead of Wi-Fi modems, Li-Fi would use transceiver-fitted LED lamps that can light a room as well as transmit and receive information. Since simple light bulbs are used, there can technically be any number of access points.

The D-Light team has developed a device that can modulate light signals to transmit and receive data. The device is being refined for commercialisation. “We have early prototypes and by

the second quarter of 2012, we'll have early Li-Fi products on the market. The initial products will be components combined with light fixtures,” shares Dr Povey. In the meanwhile, there are small problems to be overcome.

Scope for improvement

Dr Suresh Borkar is a trend-watcher, consultant and communications expert who teaches at the Illinois Institute of Technology. He opines that at the current stage of maturity, Li-Fi usage will be limited to in-house and proximity applications. The use of very high frequency (400-800 THz) limits it to very short distances and more of point-to-point communications.

The fact that Li-Fi is being considered as one of the IEEE 802.xx standards bodes well for its potential success. Like other 802.xx standards, it is defined only at layers 1 and 2 (physical and media access control (MAC) layers) of the Open Systems Interconnection (OSI) model. Layer 3 and higher layers need to be designed using the Internet Engineering Task Force (IETF) packet transport standards.

Li-Fi, according to Dr Borkar, is still in the experimental laboratory stage. Standards have to be defined